Numerical Methods for Differential Equations

Lab Project - Part III
FEM solution of a parabolic problem

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May 22, 2026

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1 Parabolic equaiton

Consider the model problem:

$$\begin{aligned} \frac{\partial u}{\partial t} &= \operatorname{div}\left[K(x)\,\nabla u\right] + f(x,t) && t \in [0,T];\ \ x \in \Omega \subset \mathbb{R}^d, \\ u(x,0) &= u_0(x) && t = 0,\ x \in \Omega, \\ u(x,t) &= u_D(x,t) && t \in [0,T],\ x \in \Gamma_D \subset \partial\Omega, \\ -K(x)\,\nabla u(x,t)\cdot\nu &= q_N(x,t) && t \in [0,T],\ x \in \Gamma_N \subset \partial\Omega. \end{aligned} \tag{1}$$

1.1 FEM method

We use the θ-method that combines explicit Euler $\theta = 0$, implicit Euler $\theta = 1$ and Crank-Nicolson $\theta = 1/2$.

$$\begin{aligned} \left(\frac{u_h^{n+1}-u_h^n}{\Delta t_n}, v\right)_{L^2(\Omega)} &+ \theta a\left(u_h^{n+1}, v\right) + (1-\theta)a\left(u_h^n, v\right) = \\ &= \theta\left(f(t_{n+1}), v\right)_{L^2(\Omega)} + (1-\theta)\left(f(t_n), v\right)_{L^2(\Omega)} \\ &\qquad\qquad \forall v \in \mathcal{V}_h \ \ n = 0, 1, \ldots, N-1, \\ \left(u_h^0, v\right)_{L^2(\Omega)} &= \left(u_0, v\right)_{L^2(\Omega)} \qquad\qquad \forall v \in \mathcal{V}_h. \end{aligned}$$

or in matrix form:

$$\left(\frac{1}{\Delta t_n}P + \theta A\right)u^{n+1} = \left(\frac{1}{\Delta t_n}P - (1-\theta)A\right)u^n + \theta b^{n+1} + (1-\theta)b^n, \tag{2}$$

where P is the mass matrix and A the stiffness matrix.

Remark 1.1.1. A better implementation, but not required for this project, is obtained when solving for the difference $\delta u = u^{n+1} - u^n$. In this case we obtain:

$$\left(\frac{1}{\Delta t_n}P + \theta A\right)u^{n+1} = -\Delta t_n A u^n + \theta b^{n+1} + (1-\theta)b^n, \tag{3}$$

where this linear system is solved with (zero) homogeneous Dirichlet boundary conditions for δu. Then the new solution is obtained as:

$$u^{n+1} = u^n + \delta u \tag{4}$$

Obviously, we assume that u^0 satisfies the real (possibly non-homogeneous) Dirichlet boundary conditions.

For $\theta = 0$ (explicit Euler) stability is ensured if:

$$\Delta t < \frac{2}{\lambda_{\max}(P^{-1}A)} \leq \frac{C}{h^2} \tag{5}$$

where $\rho(P^{-1}A)$ the specral radius of $P^{-1}A$ and C is often smaller than 1 and must be determined experimentally. No stability conditions on Δt are needed if $\theta \geq \frac{1}{2}$.

Remark 1.1.2. In a one-dimensional problem the stability condition (5) is exactly the same as for the finite difference scheme.

$$\Delta t < \frac{1}{2}\frac{h^2}{A} \tag{6}$$

where A is the value of the (scalar) diffusion coefficient.

2 Project

Solve the equation

$$\begin{aligned} \frac{\partial u}{\partial t} &= \operatorname{div}[K\,\nabla u] + f(x,t) && t \in [0,T];\ \ x \in \Omega \subset \mathbb{R}^d, \\ u(x,0) &= u_0(x) && t = 0,\ x \in \Omega, \\ u(x,t) &= 0 && t \in [0,T],\ x \in \partial\Omega, \end{aligned}$$

Use the manufactured solution $u(x,y,t) = e^{-t}\sin(\pi x)\cos(\frac{\pi}{2}y)$ and calculate the forcing function that provides the given solution, using homogeneous Dirichlet boundary conditions at all time and the initial condition $u_o(x)$.

1. write a Matlab code that implements the FEM θ method described above for eq. (**??**) using a two dimensional triangulation;
2. employ the sequence of meshes used in Project 1, possibly avoiding the use of `mesh4` and `mesh5` when the timestep Δt is too small due to stability limitations;
3. solve the problem for $T = 1$, for $\theta = 0$, $\theta = 0.5$, and $\theta = 1$;
4. calculate the $L^2(0,T;L^2(\Omega))$ norm of the error and verify that the order of convergence for all test-cases if $C_1\Delta t + C_2 h^2$ for $\theta = 1$ and $C_1\Delta t^2 + C_2 h^2$ for $\theta = 1/2$;
5. verify order of convergence and number of time steps required to reach $T_{\max}$ for $\theta = 0$.

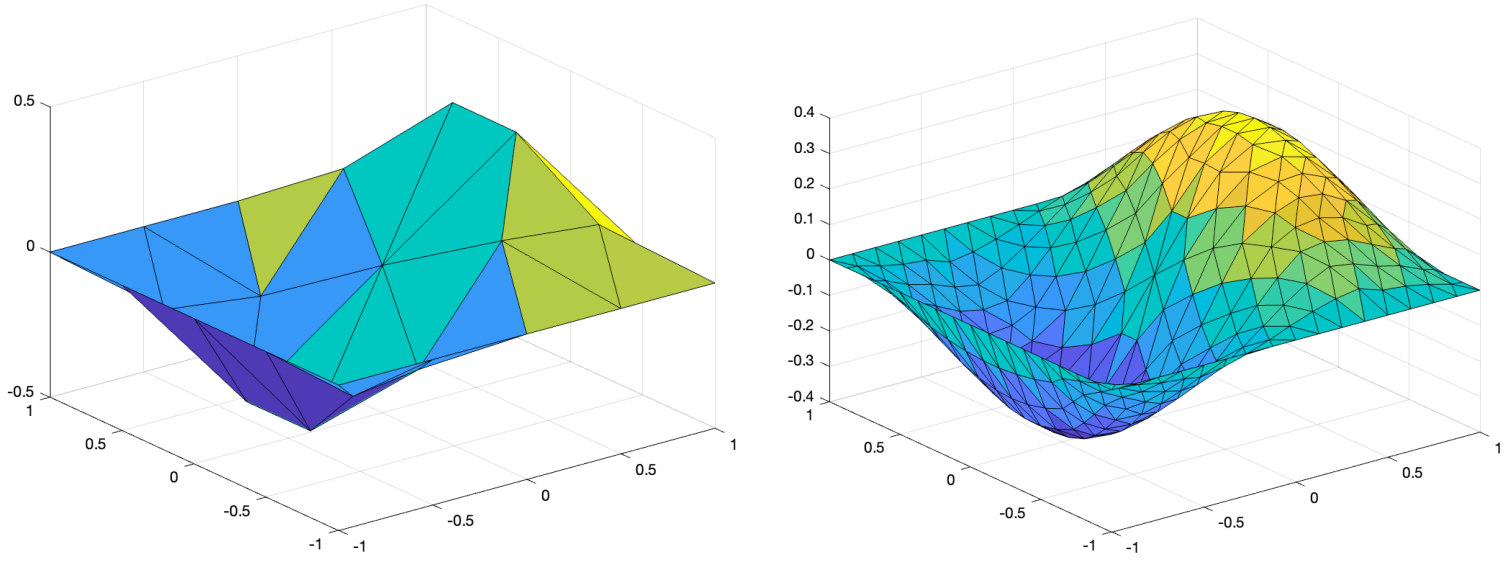


FIGURE 1: *Example of stable solutions*

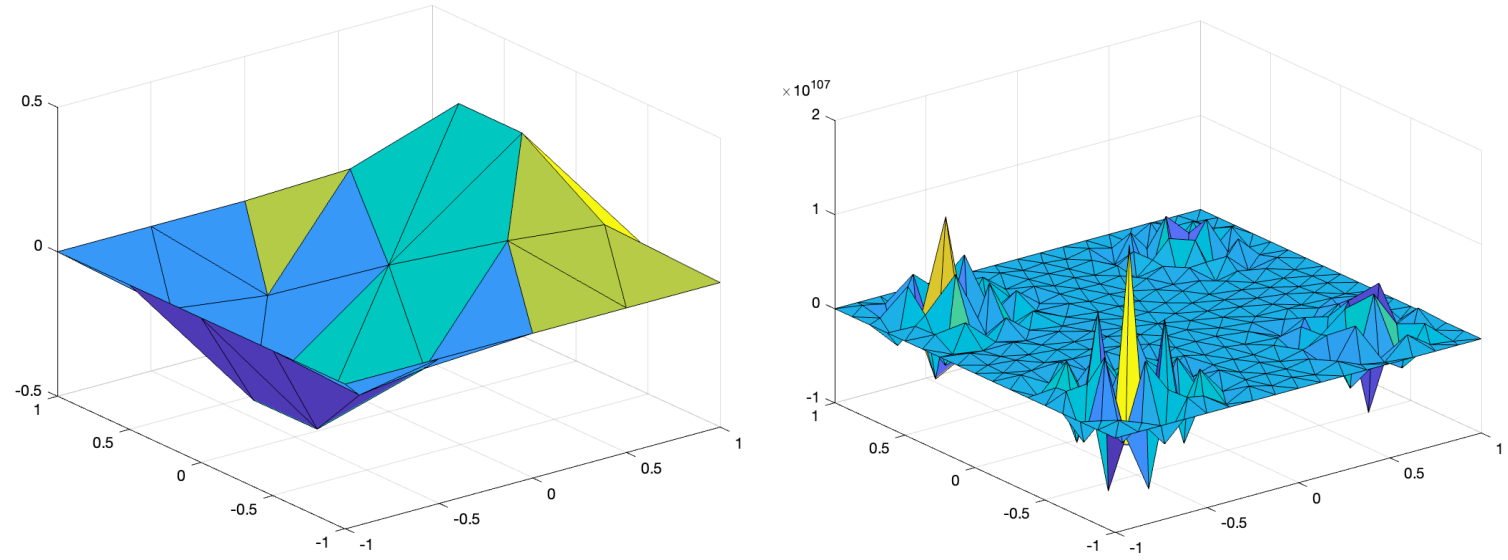


FIGURE 2: *example of unstable solutions*